Ali Alavi

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Education

The Ohio State University Columbus, OH
Ph.D. in Computer Science *Sep. 2022 – Present*
Adviser: Dr. Donald S. Williamson
Research: speech processing, neural audio coding, auditory attention detection, EEG-informed speech enhancement

Indiana University Bloomington, IN
Ph.D. in Computer Science (transferred to OSU) *Jan. 2022 – Sep. 2022*
Adviser: Dr. Donald S. Williamson

University of Tehran Tehran, Iran
M.Sc. in Communication Engineering *Sep. 2016 – Oct. 2020*
Thesis: Monte Carlo simulation of polarization-sensitive optical coherence tomography using parallel programming
Supervisor: Dr. Mahmoud Mohammad-Taheri Adviser: Dr. Mahmoud Shahabadi

Ferdowsi University of Mashhad Mashhad, Iran
B.Sc. in Electrical and Electronics Engineering *Sep. 2012 – Mar. 2016*
Capstone: LNA compensator with LabVIEW and Xmega microcontrollers
TA: Computer Programming, Logic Circuits, Computer Architecture Lab (Coordinator)

Research Experience

The Ohio State University Columbus, OH
Graduate Research Associate / GTA *Sep. 2022 – Present*

- Designed **CADENZA**, a content-adaptive neural audio codec achieving sub-kilobit-rate compression via variable-rate finite scalar quantization, conditional flow matching, and a streaming Conformer architecture; submitted to Interspeech 2026.
- Developed a contrastive-learning framework for **auditory attention detection** (AAD) from EEG that identifies the attended speaker without stimulus-specific retraining, generalizing across unseen subjects.
- Investigated self-supervised pre-training strategies for building transferable EEG representations, targeting brain–computer interface downstream tasks with limited labeled data.
- Built an EEG-informed speech enhancement pipeline that integrates decoded neural attention signals to improve noise reduction in multi-talker environments.
- Designed and built a custom multimodal data-collection setup at OSU for simultaneous recording of high-density EEG, eye-tracking (gaze), and behavioral responses during multi-talker auditory attention tasks; managed end-to-end experiment design, IRB protocol, stimulus delivery, hardware synchronization, and participant sessions to create an in-house AAD dataset with matched neural, oculomotor, and audio streams.

Indiana University Bloomington, IN
Graduate Research Assistant *Jan. 2022 – Sep. 2022*

- Initiated research on EEG-guided speech enhancement using deep learning under Dr. Donald S. Williamson.

Selected Projects

CorticalFlow — EEG-to-Audio Neural Decoding *(ongoing; targeting NeurIPS 2026)*

- Designing a multi-resolution EEG-to-audio reconstruction system that decodes perceived speech from neural signals, using an envelope-first alignment strategy that progressively reconstructs broadband envelope, mel-spectrogram, and waveform targets.
- Curating and unifying five auditory attention detection datasets (KUL, DTU, SparrKULee, NJU, AV-GC-AAD) under a standardized preprocessing and leave-one-subject-out (LOSO) evaluation protocol; systematically addressing the gaze-confound problem that inflates reported accuracy in prior AAD literature.
- Framing the core contribution around true attended-source identification (not spatial lateralization), with cross-dataset generalization benchmarks and ablations designed to meet NeurIPS 2026 standards; paper in active preparation.

AIMO3 — RL for Mathematical Reasoning in LLMs

- Developed a complete strategy for the AIMO3 Kaggle competition (AI Mathematical Olympiad) centered on Group Relative Policy Optimization (GRPO) and reinforcement learning from verifiable rewards (RLVR) applied to open-weight math-reasoning LLMs (Qwen2.5-Math, DeepSeek-R1 distillations).
- Implemented inference-time compute scaling with Best-of-N sampling, majority voting, and Monte Carlo Tree Search (MCTS) with Lean 4 formal verification for proof-level correctness guarantees.

Self-Supervised Learning & Vision Foundation Models

- Conducted in-depth study of Joint Embedding Predictive Architectures (JEPA/V-JEPA), DINOv2, SigLIP-2, MAE, VICReg, and Barlow Twins with full mathematical derivations of loss functions and convergence properties.
- Applied self-supervised frameworks to the biosignal domain, adapting contrastive and predictive objectives for EEG representation learning under limited labeled data and high inter-subject variability.

Industry Experience

BNtelligence Columbus, OH
Computer Vision Engineer *Jul. 2025 – Aug. 2025*

- Built an end-to-end defect-detection pipeline: collected and annotated visual data, trained classification and semantic segmentation models, and deployed a real-time video inference API serving production traffic.

AI-bridge Hamburg, Germany
Deep Learning Consultant *Oct. 2019 – Mar. 2020*

- Designed and shipped production deep learning applications—image enhancement, body-part segmentation, and super-resolution—using TensorFlow and PyTorch.

Payafanavaran Mashhad, Iran
Software Engineer *Jan. 2019 – Aug. 2019*

- Developed a C# application for 3D surface reconstruction from structured-light scanning data.

Tarashe Sanat-e Pishro Mashhad, Iran
Hardware Design Engineer *Jul. 2016 – Sep. 2016*

- Designed a high-speed FPGA board integrating DDR3 SDRAM, Ethernet, and multiple peripherals for industrial data acquisition.

Publications

1. **S. A. Alavi Bajestan**, D. S. Williamson, "NEUROTOKEN: Joint Source-AAD, Direction-AAD and Envelope Decoding on Short EEG Windows via Conditional Flow Matching," *Submitted to Neurips*, 2026.
2. **S. A. Alavi Bajestan**, H. Naimul, D. S. Williamson, "Beyond EEG: MAESTRO, a Multimodal Dataset for Decoding Auditory Attention from Eye, Brain, and Body," *Submitting to IEEE TASLP*, 2026.
3. **S. A. Alavi Bajestan**, D. S. Williamson, "CADENZA: Content-Adaptive Encoding with Zero-Collapse Quantization for Sub-Kilobit Neural Audio Compression," *Submitted to Interspeech*, 2026.
4. **S. A. Alavi Bajestan**, M. Pitt, D. S. Williamson, "A Contrastive-Learning Approach for Auditory Attention Detection," *arxiv.*
5. **S. A. Alavi Bajestan**, D. S. Williamson, "Contrastive Learning Approach for Source Identification in Auditory Attention Detection," *Poster at Midwest Speech and Language Days*, 2024.
6. **S. A. Alavi Bajestan**, "Self-Supervised Learning for Auditory Attention Detection," *Poster at Center for Cognitive and Brain Sciences, The Ohio State University*, 2023.
7. **S. A. Alavi Bajestan**, "Neural Network for Estimation of Optical Characteristics of Optically Active and Turbid Scattering Media Using Monte Carlo Simulation," *arXiv:2011.06934*, 2020.

Technical Skills

Programming: Python, C/C++, C#, CUDA, MATLAB, Java, JavaScript

ML Frameworks: PyTorch, TensorFlow, PyTorch Lightning, scikit-learn

Research Areas: speech enhancement, neural audio coding, auditory attention detection, EEG/BCI, foundation models, reinforcement learning for LLMs (GRPO/PPO/RLVR), self-supervised learning (JEPA, contrastive, masked), generative modeling (VAE, GAN, diffusion, flow matching), computer vision, Koopman operator theory, transformer and mixture-of-experts architectures

Tools & Infrastructure: Git, Linux, Docker, HPC/Slurm, Weights & Biases, Hydra, LaTeX, Lean 4

Hardware: FPGA (VHDL/Verilog), ARM Cortex-M, LabVIEW, Cadence IC Design

Honors & Service

- CCBS Summer Research Award, The Ohio State University, 2023
- 3rd Place, CMDC Hackathon
- Reviewer: EACL 2021 – AfricaNLP Workshop
- Student Volunteer: ICLR 2021, ICML 2021
- Student Volunteer: ICEE Conference, Ferdowsi University of Mashhad, 2015
- Coordinator, Entrepreneurship Center, Ferdowsi University of Mashhad, 2015–2016